

Section-A (25 Marks)

1. What are the technologies available to determine best route location of highways?

Answer:

The best route location of highway is determined using modern surveying, mapping, and geospatial technologies. These technologies help to identify the most economical, safe, and environmentally suitable alignment.



The technologies available to determine the best route location of highways are:

1. Remote Sensing

- It uses satellite imagery and aerial photographs.
- Remote sensing helps to identify land use, vegetation, rivers, settlements, and geological features.
- It is useful for preliminary route selection over large areas.

2. Geographic Information System (GIS)

- GIS integrates spatial and attribute data to analyze multiple route alternatives.
- It considers slope, soil, geology, drainage, land use, environmental constraints, and cost.
- It is widely used for **Multi-Criteria Decision Analysis (MCDA)** in route optimization.

3. Global Navigation Satellite System (GNSS) / GPS

- GPS system provides accurate geographic coordinates and elevation.
- It is used for reconnaissance surveys, alignment marking, and mapping.

4. LiDAR (Light Detection and Ranging)

- It uses laser pulses from aircraft or drones to generate highly accurate 3D terrain models.
- LiDAR is effective in dense forests and mountainous terrain.
- It helps to produce Digital Elevation Models (DEM) and contour maps.

5. Drone (UAV) Surveying

- This method captures high-resolution aerial images quickly and economically.
- Drone surveying is very useful for inaccessible or rugged areas.
- Generates orthophotos and 3D terrain models.

6. Photogrammetry

- Photogrammetry creates accurate maps and 3D models from overlapping aerial photographs.
- It is used for topographic surveys and corridor mapping.

7. Total Station Survey

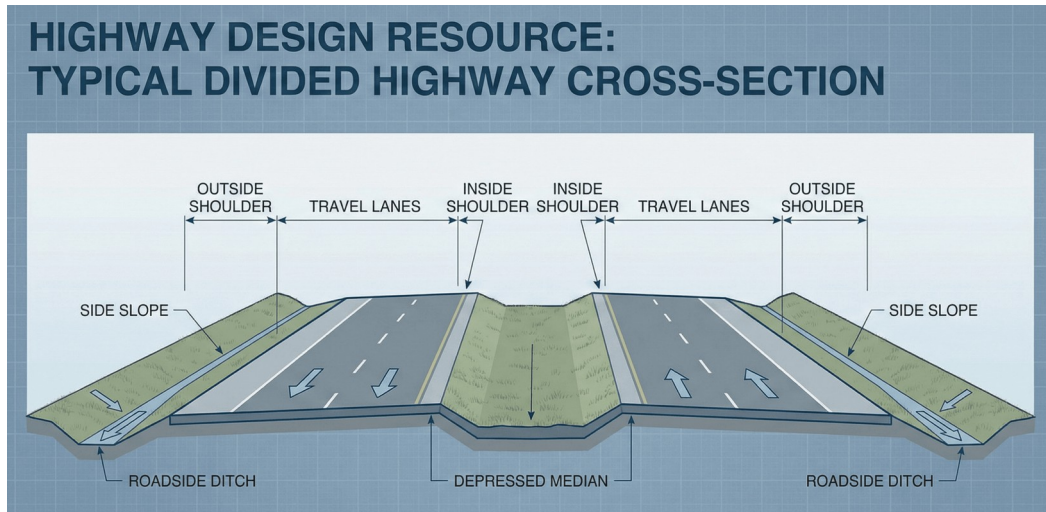
- TS is an electronic instrument for precise measurement of horizontal and vertical angles and distances.
- It is used during detailed alignment surveys and setting out.

2. a. What are the basic factors which influences the visible dimensions of a road?

b. What correction is required in horizontal curves to negotiate the speed and stability?

Answer:

a. The visible dimensions of a roads are influenced by several critical engineering and environmental factors. These elements include the carriageway (pavement width), shoulders, roadway width, and side slopes.



Based on the sources, the primary factors influencing these dimensions are:

i. Design speed:

- Design speed is considered the most critical factor affecting the geometric design of a highway.
- It determines the sight distance, the radius of horizontal curves, and the length of vertical curves.
- Higher design speeds require larger radii of curvature and more generous dimensions for safety and comfort.

ii. Design vehicle:

The dimensions, weight of axles, and operating characteristics of the design vehicle (such as a standard truck or passenger car) directly influence road dimensions.

- **Pavement Width:** The width of traffic lanes is determined by the width of the design vehicle. For example, standard lane widths in Nepal range from 3.5 m to 3.75 m depending on the road class.
- **Extra-Widening:** On sharp curves, additional width (mechanical and psychological widening) is required to accommodate the "off-tracking" of longer vehicles.
- **Clearance Distances:** Overhead and lateral clearances must accommodate the maximum height and width of expected vehicles.

iii. Topography:

The physical layout and "ruggedness" of the land dictate what dimensions are feasible and economical.

- Terrain is classified into plain, rolling, mountainous, or steep based on the cross slope.
- In steep mountainous terrain (cross slopes > 60%), design speeds are reduced to minimize massive excavation, which results in narrower curve radii and more restricted visible dimensions compared to plain terrain.

iv. Traffic volume and capacity:

The amount of traffic a road is expected to carry determines its total width.

- **Number of Lanes:** Engineers use the 30th highest hourly volume to determine the required number of traffic lanes.
- **Roadway Width:** This is the sum of the width of pavements, separators, and shoulders. High-capacity roads require wider formations to include medians and traffic separators.

v. Safety and functional requirements:

Sight Distance: The dimensions must ensure a driver can see stationary or moving objects far enough ahead to stop safely (Stopping Sight Distance) or pass other vehicles (Overtaking Sight Distance).

Shoulders: These are provided along the road edge for stopped vehicles and lateral support. Standard practice in Nepal suggests a minimum width of 0.75 m on either side.

Right of Way (ROW): This is the total width of land acquired for the road, which must be adequate to house all cross-sectional elements and allow for future development.

b. To ensure safe negotiation of horizontal curves at the design speed and maintain vehicle stability, the following corrections (design provisions) are provided:

i. **Superelevation:**

- The outer edge of the pavement is raised above the inner edge, which counteracts the centrifugal force acting on vehicles while negotiating the curve is known as superelevation.
- It improves stability and reduces the possibility of skidding and overturning.
- IRC/DOR generally limit maximum superelevation to 7–10% depending on terrain and climate.

ii. **Extra Widening**

- Extra widening is an additional pavement width is provided on curves.
- Compensates for:
 - i. **Mechanical widening** due to off-tracking of rear wheels.
 - ii. **Psychological widening** because drivers require extra clearance at higher speeds.
- Particularly necessary on sharp curves with small radii.

iii. **Transition Curves**

- Spiral curves are introduced between the tangent and circular curve.
- They provide a gradual change in curvature and allow:
 - i. Smooth introduction of centrifugal force.
 - ii. Gradual development of superelevation.
 - iii. Improved riding comfort and safety.

iv. **Adequate Sight Distance**

- Obstructions on the inside of the curve are removed.
- It ensures the required Stopping Sight Distance (SSD) and Overtaking Sight Distance (OSD) where applicable.

v. **Radius of Curve**

- A larger radius reduces centrifugal force.
- Radius should satisfy the design relationship:

$$e + f = \frac{V^2}{127 R}$$

where :

e = super elevation,

f = coefficient of lateral friction,

V = design speed (km/h),

R = radius of curve (m).

vi. **Proper Pavement Surface and Drainage**

- Good skid resistance and effective drainage prevent skidding, especially during rainfall.

3. **It is argued that road construction in hills is one of the major causes of natural environmental degradation. Examine how the road construction project in hills effect the environment? What measures would you suggest to take up at various stages of the project for minimizing the environmental damages?**

Answer:

Road construction in hills is one of the major causes of natural environmental degradation. Road construction in hilly regions, particularly in the geologically fragile terrain of the Himalayas and Middle Hills of Nepal, is a double-edged sword. While it provides essential connectivity, it is also a primary driver of natural environmental degradation.

➤ **Slope Instability and Landslides:**

Excavation and excessive cutting of hillslopes for road alignments often trigger landslides and rockfalls. Human intervention like removing natural toe support or blasting makes soil mass loose, causing it to slide under gravity.

➤ **Soil Erosion and Weathering:**

Newly cut slopes and embankments are highly susceptible to the direct impact of rainfall and high-speed surface runoff. Without protection, this leads to gully erosion, the formation of cross ruts, and the washing away of topsoil.

➤ **Deforestation and Biodiversity Loss:**

Road alignment through forested areas requires clearing vegetation, which removes the natural anchorage and reinforcement provided by tree roots. This leads to habitat destruction due to deforestation.

➤ **Hydrological Disruption:**

Improper drainage management allows water to percolate into the subgrade, increasing pore water pressure which is one of the major causes of slope failure. Furthermore, construction can alter natural stream paths and increase downstream sedimentation.

To minimize these environmental damages, environmental safeguards must be integrated into every stage of the project lifecycle:

i. **Planning stage:**

- Conduction of mandatory Environmental Impact Assessments (EIA) or Initial Environmental Examinations (IEE) to identify sensitive ecosystems, landslide-prone zones, and cultural heritage sites before construction begins.
- Roads should be aligned as closely as possible to natural ground contours to minimize massive excavation and filling operations.
- Alignment should be shifted away from active or old landslide zones, faults, and shear zones.
- Engineers must identify obligatory points to avoid, such as religious places, costly existing structures, or marshy lands that require excessive compensation or stabilization.

ii. **Design stage:**

- Design of for vegetative stabilization (e.g., turfing, fascines, palisades) alongside civil structures. This is more economical, sustainable, and aesthetically pleasing than heavy concrete walls alone.
- Design of efficient side drains, catch-water drains, and subsurface horizontal drains to intercept runoff and prevent water from weakening the road subgrade or the slope.
- Provision of structures like aqueducts in steep areas where traditional culverts are not feasible to maintain natural water flow.

iii. Construction stage:

- Avoid excessive cutting and the use of heavy machinery in geologically unstable areas where vibrations might trigger a slide.
- A balanced excavation is preferred between cut and fill volumes to reduce the need for external borrow pits or the disposal of excess debris into rivers.
- Implementation of standards to regulate dust, smoke, and noise generated by construction machinery.

iv. Operation and maintenance stage:

- Regular clearing of debris from side drains and culverts, especially before and during the monsoon, to prevent water-induced damage to the road and surrounding slopes.
- Maintain the growth of bio-engineering plants to ensure they remain strong and effective as natural soil reinforcement over time.
- Conduct regular inspections to identify early signs of distress, such as small cracks or ruts, which can be repaired before they escalate into major environmental disasters.

4. a. Explain the post card methods of O and D surveys. Discuss the advantage and disadvantage of this methods?

b. How are the road intersection planned? Enumerate the various traffic controls needed at an intersection.

Answer:

a. O and D surveys is carried out to collect information about the pattern of journeys that people make, amount of travel between various locations and so on. There are several methods of conducting O and D surveys. Among them, Post Card Method is an indirect method of conducting an Origin and Destination (O–D) survey.

In this method, postcards or survey forms are distributed to drivers at selected survey stations. The drivers fill in trip information and return the completed card by post.

- Survey personnel stop vehicles at selected locations.
- A pre-addressed and postage-paid postcard (or survey form) is given to the driver.
- The driver fills in the required information after completing the trip, such as:
 - Origin of the trip
 - Destination of the trip
 - Route followed
 - Purpose of travel
 - Type of vehicle
 - Date and time of travel
- The completed postcard is mailed back to the survey agency.
- The collected responses are analyzed to determine travel patterns and traffic characteristics.

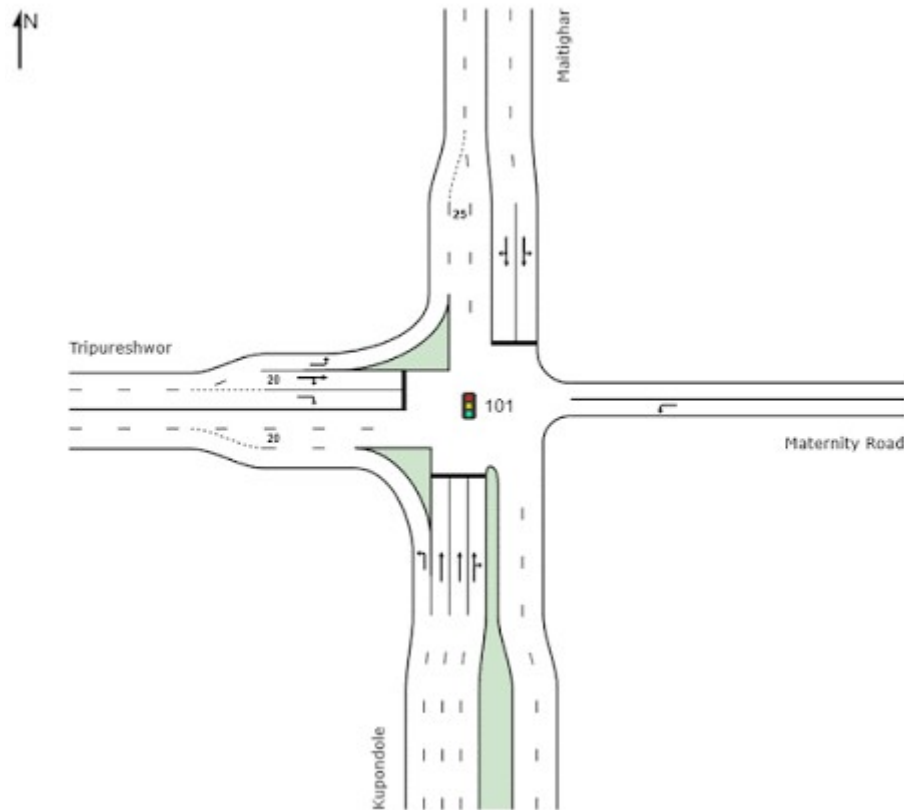
Advantages:

- Cheaper to conduct this survey than home-interview or roadside surveys.
- Easy to distribute postcards at survey stations.
- Fewer survey staffs are required in comparison to direct interview methods.

Disadvantages:

- Results are delayed because forms are returned by post.
- Surveyors cannot verify or explain questions once the postcard is handed out.
- Due to low response and possible errors, it is less reliable than direct interview methods.

b. Road intersection is defined as an area where two or more roads join or cross within which are included with the roadway and roadside facilities for traffic movement in that area. It is desirable that the intersection roads meet at or nearly at right angles.



Road intersections are planned considering the following points:

- i. There should be adequate visibility for all the approaching vehicles on all converging roads.
- ii. Width of pavement at radius for turning should be sufficiently provided.
- iii. Area of conflict should be reduced to minimum by introducing properly designed traffic island and channelizing island.
- iv. Sudden change of path should be provided.

Various devices that are used to control, regulate and guide traffic are known as traffic control devices. These devices provide safe and efficient functioning of intersections by separating conflicting vehicle streams in time. There are different types of traffic control devices. They are:

i. Traffic sign:

- Traffic sign conveys the specific information to driver in advance that allows him/her to be careful.
- It gives timely warning of hazardous situations and information of routes, directions and intersection in the highway.
- Traffic signs can be regulatory, warning signs or informatory signs.

ii. Traffic signals:

- Traffic signals are such control devices which alternatively direct the traffic to stop and process at level intersections.
- The main requirement of traffic signals is to draw attention, provide information and enough time to respond to the drivers.
- Traffic signals can be traffic control signals or pedestrian signal or special signal.

iii. Traffic island:

- Traffic island is the raised area provided at the level road to avoid or minimize the areas of major and minor conflicts.
- Island establishes physical channels through which vehicular traffic may be guided to travel and avoid conflict with other vehicular traffic.

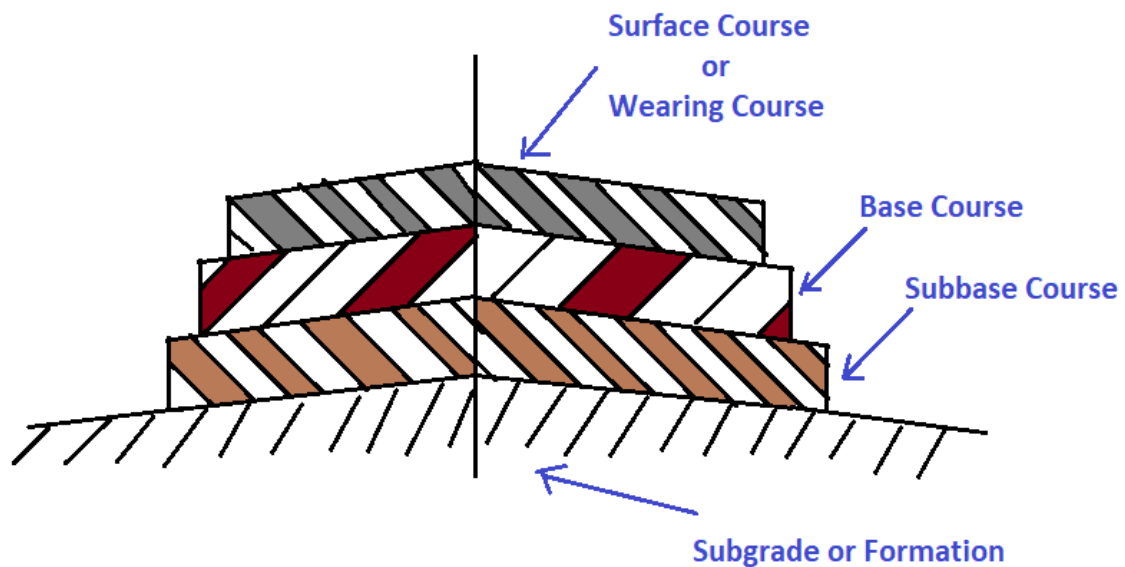
iv. Road markings:

- It is used as means of controlling and guiding traffic.
- Traffic markings may be in the form of line pattern, words, symbols or reflectors on the pavement.
- Road markings promote smooth and harmonious flow of traffic along the guided paths of travel.

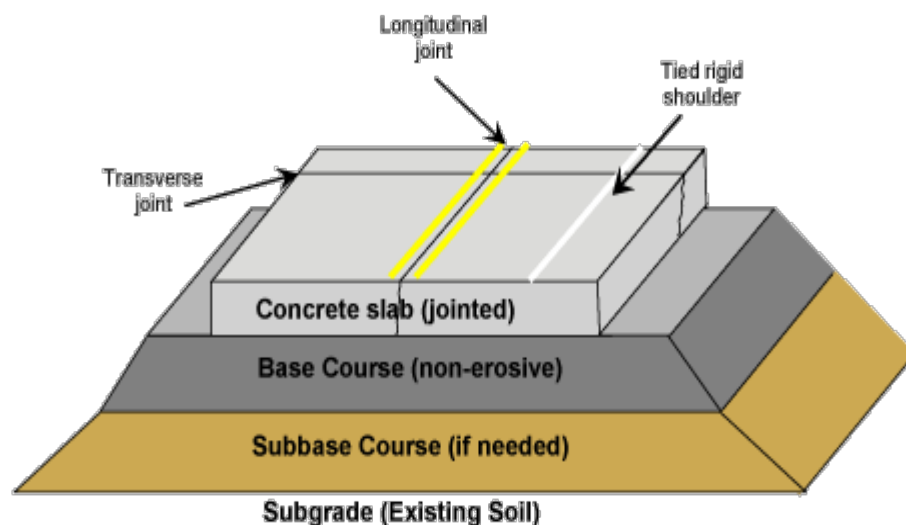
5. What are flexible and rigid pavements? Describe them with sketch. Which types of pavements is suitable for hill roads in Nepal? Give reasons.

Answer:

The pavements which have very low structural strength and are flexible in their structural behavior under the load are called flexible pavements. The flexible pavements layers reflect the deformation of the lower layers onto the surface of the layer by grain-to-grain contact. The strength of the pavement depends on the quality and thickness of each layer.



Rigid pavements are those which possess considerable flexural strength, made up of cement concrete which may either be plain, reinforced or prestressed. They have a slab action and are capable of transmitting the loads stresses through a wider area below.



In the context of Nepal's hilly terrain, flexible pavements are generally considered more suitable for the following reasons:

1. **Adaptability to Geological Instability:**

The hills of Nepal are geologically fragile and prone to failure. Flexible pavements can adjust to subgrade irregularities and differential settlements without catastrophic failure, whereas rigid pavements would rupture.

2. **Economic Viability:**

Hill roads are expensive to construct due to steep slopes and extensive earthworks. Flexible pavements have a lower initial construction cost, making them more feasible for expanding road networks in difficult terrain.

3. **Ease of Maintenance:**

Maintenance in remote hilly regions is challenging. Flexible pavements allow for easy localized repairs (such as patchwork for potholes) compared to the complex joint repairs or slab replacements required for rigid pavements.

4. **Immediate Operation:**

Once constructed and cooled, flexible roads can be opened to traffic immediately, which is vital for maintaining connectivity in regions where alternate routes are often non-existent. Rigid pavements require at least 14 days of curing before use.

6. **a. How planning for road maintenance operations is carried out?**

b. Why is hydraulic analysis necessary for river bank and protection structures?

Answer:

a. The systematic process of identifying road defects, prioritizing maintenance needs, allocating resources, and scheduling maintenance activities to keep roads safe, serviceable, and economical throughout their design life is known as road maintenance planning.

Planning for road maintenance operations is carried out in following steps:

i. Road inventory:

- The information regarding road length, width, pavement type. Drainage structures, cross drainage structures, traffic volume are collected.

ii. Road condition survey:

- The defects in the pavements such as potholes, cracks, rutting, corrugations, edge failures etc. are inspected and surveyed the condition of road.

iii. Assessment and prioritization:

- The severity and urgencies of defects are evaluated.
- The roads are prioritized based on traffic volume, road conditions, safety concerns etc.

iv. Selection of maintenance type:

- The appropriate maintenance operation is chosen:
 - ✓ **Routine maintenance:** Cleaning drains, patching potholes, vegetation control.
 - ✓ **Periodic maintenance:** Resealing, overlay, surface renewal.
 - ✓ **Emergency maintenance:** Flood, landslide, and accident damage repairs.
 - ✓ **Rehabilitation:** Major repairs to restore pavement strength.

v. Resource planning and work scheduling:

- Labor, construction materials, equipment and machinery, budget and funding are estimated and arranged.

- Annual, seasonal, monthly maintenance plans are prepared.
- In Nepal, major maintenance is generally scheduled before and after the monsoon.

vi. Implementation:

- Maintenance must be carried out according to specifications.
- Traffic management and worker safety during construction.

vii. Monitoring and evaluation:

- Completed works must be properly monitored.
- The quality and performance must be inspected.

b. Hydraulic analysis is a fundamental requirement for the design and construction of river bank and protection structures as it ensures that these engineering interventions can withstand the dynamic forces of a river while maintaining long-term stability.

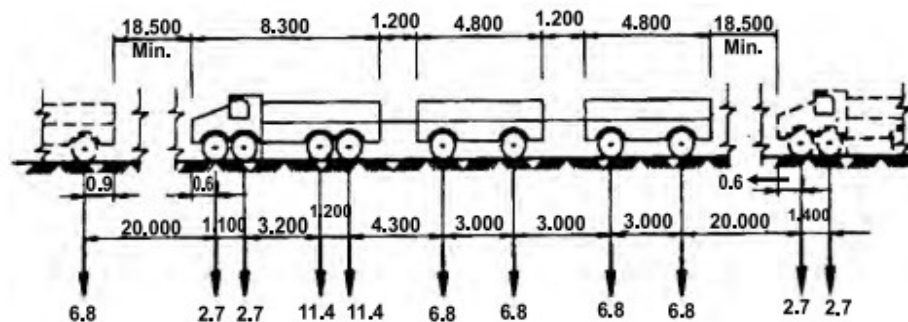
Hydraulic analysis necessary for river bank and protection structures because:

- It helps in determination of design flood which ensures protection of structure from extreme flood events.
- Hydraulic analysis helps in assessment of flow velocity which enables selection of suitable materials such as ripraps, gabions or concrete blocks.
- It also helps in estimation of scour depth and depth of foundations.
- Hydraulic analysis enables prediction of flood levels and fixes the required height of embankments, levees and flood walls.
- It helps to prevent siltation, deposition patterns and bank erosion.
- It evaluates hydraulic forces acting on the structures.
- It also helps to minimize the adverse impacts on the river ecosystem and maintains natural flow patterns and aquatic habitats.

7. Draw a detail configuration of live load of IRC class A bridge loading for a design of a single lane R.C.C. bridge.

Answer:

- IRC Class A loading is the standard live load specified by the Indian Roads Congress (IRC), commonly adopted in Nepal for the design of highway bridges carrying normal traffic.



Class A Train of Vehicles

- It consists of driving vehicle, trailers of specified axle, spacing and loads.
- IRC class A loading is adopted on all the roads where permanent bridges are constructed.

- Bridges designed for IRC Class A loading should also be designed for IRC class AA loading or 70 R loading.
- Total length of loading is about 25 m, with the total load of 32.3t.
- The nose to tail spacing between two successive trains shall not be less than 18.5 m.
- For a single-lane bridge, the carriageway width is generally 3.75 m, and the IRC Class A vehicle is placed centrally to produce the maximum bending moment and shear force.
- It is the most commonly used loading for single-lane RCC highway bridge design.
- For heavily loaded or industrial routes, IRC Class 70R loading is used instead of Class A.

8. **What is vegetation structure? Describe with sketches vegetative engineering techniques for slope stabilization.**

Answer:

Vegetation structure is the arrangement, distribution, and organization of plants such as grasses, shrubs, and trees on a slope. In slope stabilization, vegetation acts as a natural reinforcement that binds the soil particles with roots, reducing runoff velocity, increasing infiltration, and protecting the soil from erosion.

Different vegetative engineering techniques for slope stabilization are as follows:

i. **Grass turfing:**

- Grass is planted or seeded on exposed slopes.
- Dense fibrous roots bind the topsoil.
- Suitable for gentle slopes (up to about 30°).
- Controls surface erosion caused by rainfall.



ii. **Tree plantation:**

- Deep-rooted shrubs and trees reinforce deeper soil layers.
- Reduce landslides and improve long-term slope stability.
- Common species in Nepal include Vetiver grass, Broom grass (Amriso), Bamboo, Salix (Willow), and Alder (Utis).



iii. Live fascines:

- Bundles of live branches are tied together and laid along contour lines.
- Reduce runoff velocity and trap sediments.



iv. Vegetative gabions/ Crib wall:

- Stone-filled gabions or timber crib walls are combined with vegetation.
- Provide immediate structural support and long-term biological reinforcement.



9. Compute the intensities of active and passive earth pressure at depth of 8 meters in dry cohesionless sand and angle of internal friction of 30 degrees and unit weight of 1.8 t/m³. What will be intensities of active and passive earth pressure if the water level rises to the ground level? Take saturated unit weight of sand as 2.2 t/m³.

Answer:

Given,

$$\text{Depth, } h = 8 \text{ m}$$

$$\text{Angle of internal friction, } \phi = 30^\circ$$

$$\text{Dry unit weight, } \gamma = 1.8 \text{ t/m}^3$$

$$\text{Saturated unit weight, } \gamma_{sat} = 2.2 \text{ t/m}^3$$

$$\text{Unit weight of water, } \gamma_w = 1.0 \text{ t/m}^3$$

$$K_a = \frac{1 - \sin \phi}{1 + \sin \phi}$$

$$K_p = \frac{1 + \sin \phi}{1 - \sin \phi}$$

Since,

$$\sin 30^\circ = 0.5$$

$$K_a = \frac{1 - 0.5}{1 + 0.5} = \frac{0.5}{1.5} = 0.333$$

$$K_p = \frac{1 + 0.5}{1 - 0.5} = \frac{1.5}{0.5} = 3$$

a. Dry Sand

Active Earth Pressure Intensity:

$$\begin{aligned} \sigma_a &= K_a \gamma h \\ &= 0.333 \times 1.8 \times 8 = 4.8 \text{ t/m}^2 \end{aligned}$$

Passive earth pressure intensity:

$$\begin{aligned} \sigma_p &= K_p \gamma h \\ &= 3 \times 1.8 \times 8 = 43.2 \text{ t/m}^2 \end{aligned}$$

b. Water Table at ground level:

i. Submerged unit weight:

$$\begin{aligned} \gamma' &= \gamma_{sat} - \gamma_w \\ &= 2.2 - 1.0 = 1.2 \text{ t/m}^3 \end{aligned}$$

ii. Effective earth pressure:

Active soil:

$$\sigma'_a = K_a \gamma' h$$
$$= 0.333 \times 1.2 \times 8 = 3.2 \text{ t/m}^2$$

Passive soil:

$$\sigma'_p = K_p \gamma' h$$
$$= 3 \times 1.2 \times 8 = 28.8 \text{ t/m}^2$$

iii. Water pressure:

Hydrostatic pressure at 8 m depth:

$$u = \gamma_w h = 1 \times 8 = 8 \text{ t/m}^2$$

iv. Total earth pressure intensities:

Total active pressure:

$$\sigma_{a(\text{total})} = 3.2 + 8 = 11.2 \text{ t/m}^2$$

Total Passive Pressure

$$\sigma_{p(\text{total})} = 28.8 + 8 = 36.8 \text{ t/m}^2$$

10. What is difference between pile and well foundation in bridges? Explain with sketches.

Answer:

The difference between pile and well foundation in bridges:

Pile foundation	Well foundation
i. Pile foundation consists of slender structural members called piles driven or bored into the ground.	i. Well foundation consists of large hollow masonry or RCC caisson sunk to the required depth.
ii. Load is transferred either through end bearing, skin friction or both of the piles.	ii. Load is transferred through base bearing and side friction in well foundation.
iii. Pile foundation is constructed by driving the piles or boring or cast-in-situ using piling equipment.	iii. Well foundation is constructed at ground level and sunk by dredging up to the required depth.
iv. It is suitable in soft soils, moderate loads and restricted working space.	iv. It is suitable for the large bridge piers and bridge subjected to heavy loads.
v. Pile foundation is less suitable where scour is severe, unless long piles are provided.	v. Well foundation is highly suitable for deep scour and fast-flowing rivers.
vi. The cost of construction of pile foundation is economical for small and medium bridges.	vi. The cost of construction for well foundation is expensive but economical for major river bridges.
vii. The lateral load resistance of pile foundation is moderate.	vii. The lateral load resistance of pile foundation is excellent.

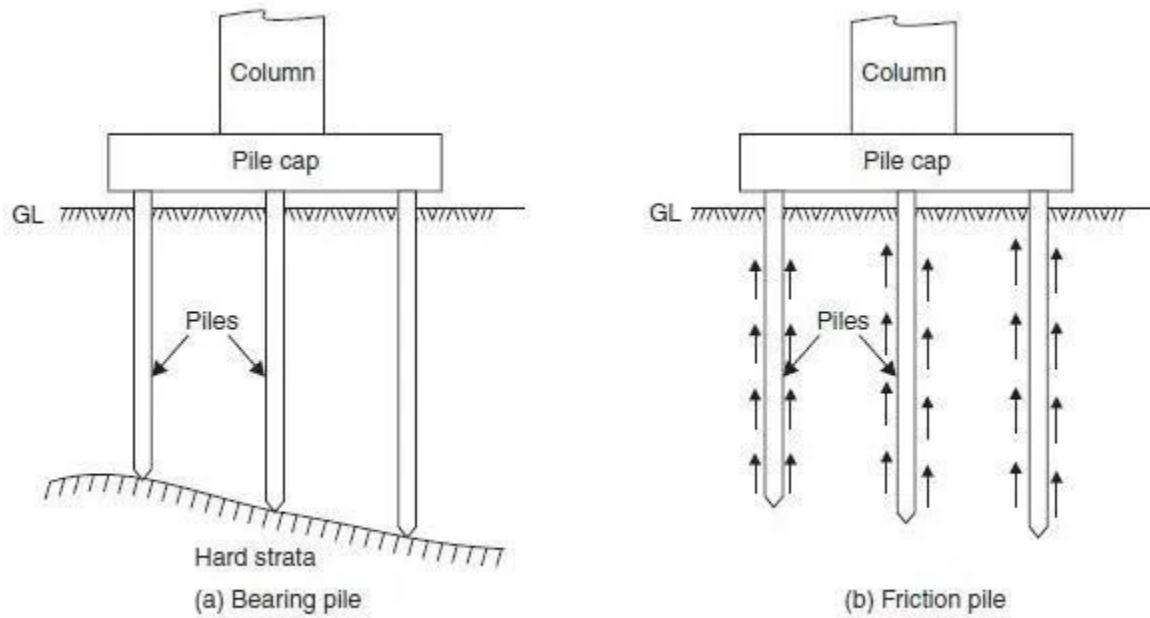


Fig. 7.9. Pile foundations

